

Smart Billing System – Project Report

Name – Harsh Rajesh Botre

1. Abstract

This report presents a comprehensive overview of a Smart Billing System designed to automate and simplify the checkout process in retail environments. The system integrates several essential technologies including a barcode scanner for identifying products, a load cell combined with the HX711 module for accurate weight verification, and a QR-based UPI payment interface for quick and contactless transactions.

The Smart Billing System enhances customer convenience by reducing human intervention, speeding up the checkout process, and eliminating long queues. It also addresses security concerns through real-time anti-theft mechanisms by verifying scanned items against their expected weight. Additionally, the system allows users to cancel mistakenly scanned products, ensuring accuracy in the billing process.

By employing the ESP32 microcontroller for processing and wireless communication, and using compact and low-power components, the system is portable, scalable, and cost-effective. It is especially suitable for supermarkets, self-service stores, and retail outlets aiming for efficient and autonomous checkout solutions.

2. Introduction

Retail operations often face numerous challenges such as prolonged billing queues, frequent manual entry mistakes, and overall customer dissatisfaction due to delays and inefficiencies. Traditional billing methods require human intervention for scanning, totaling, and payment processes, which not only increase labor costs but are also prone to human error and fraud.

These outdated systems also struggle with scalability and often fail to keep up with the expectations of modern consumers who demand speed, accuracy, and convenience. Furthermore, existing billing setups lack provisions for real-time item verification or anti-theft mechanisms, making them less reliable in high-traffic retail settings.

This project introduces a Smart Billing System as a viable solution to these challenges. By leveraging embedded systems, real-time data processing, and automated item verification, the system streamlines the checkout process. It not only reduces transaction time but also enhances accuracy and customer satisfaction by eliminating manual billing errors. Additionally, it promotes a seamless and self-service shopping experience, aligning with the current trends in digital retail transformation.

3. Literature Review / Existing Systems

Current billing systems predominantly depend on cashier-operated barcode scanners and manual data entry, which not only slow down the checkout process but also make the system prone to human error. These methods often result in customer dissatisfaction, especially during peak shopping hours. While automated systems like RFID-based smart carts have been proposed, their high cost, complexity, and infrastructure requirements limit widespread adoption.

Additionally, most conventional billing systems do not support dynamic features such as product cancellation after scanning, real-time validation of product weight for anti-theft measures, or autonomous operation without staff supervision. These limitations highlight the need for a more adaptable and intelligent solution.

This project addresses these shortcomings by introducing a cost-effective, self-service Smart Billing System. It empowers customers to manage their own billing process through barcode scanning, integrated weight validation, and instant payment via QR codes. The goal is to make the retail checkout experience faster, more reliable, and largely independent of manual intervention.

4. Proposed System

The proposed Smart Billing System integrates all necessary components on a smart shopping cart or kiosk. It reads product barcodes, verifies item weights to detect discrepancies (anti-theft), maintains a running total, and enables checkout through a dynamic QR code.

Core Functionalities:

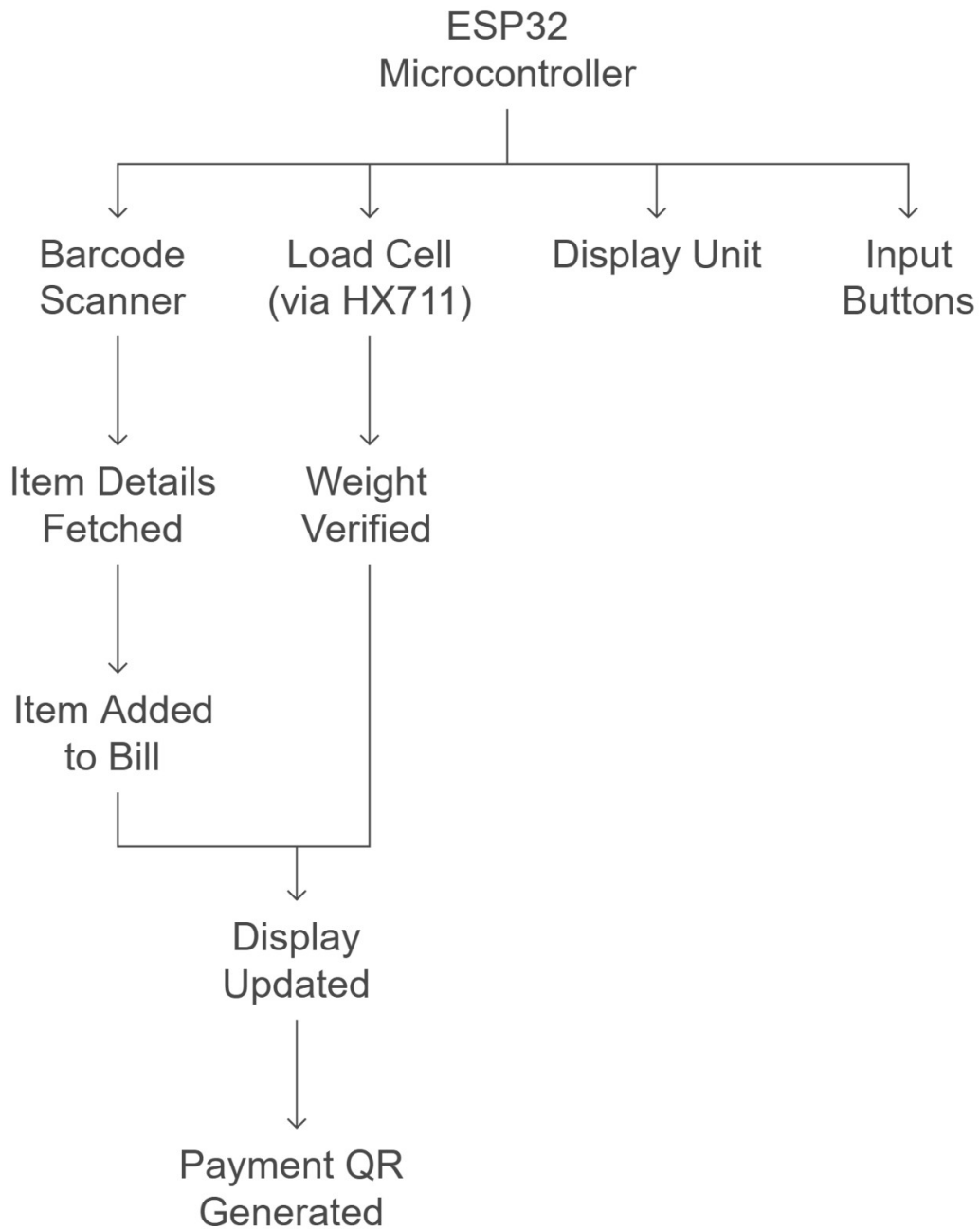
- Barcode-based item recognition
 - Real-time billing updates
 - Weight verification using load cell
 - Manual item cancellation option
 - QR-based UPI payment
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5. System Architecture

The system architecture centers around the ESP32 microcontroller, which acts as the processing hub for all input/output devices.

- Block Diagram: Shows interconnection between barcode scanner, load cell (via HX711), display unit, and input buttons.
- Data Flow:
 - Barcode scanned → Item details fetched → Item added to bill → Weight verified → Display updated → Payment QR generated

System Architecture and Data Flow



6. Hardware Components

- ESP32 Dev Board: Handles logic, input/output, and Wi-Fi communication.
 - Barcode Scanner (e.g., DE2120): Scans product barcodes to retrieve item information.
 - HX711 + Load Cell: Measures item weight and verifies it against stored values to detect mismatches.
 - OLED/TFT Display: Displays item names, prices, and total bill.
 - Push Buttons: Enable item cancellation and checkout operations.
 - Power Supply: Boost converter and rechargeable battery system for stable power delivery.
 - Buzzer (Optional): Provides audio alerts for invalid scans or errors.
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7. Software Requirements

- Arduino IDE: For coding and uploading firmware to the ESP32.
 - Libraries Used:
 - HX711.h for load cell interface
 - U8g2lib.h or Adafruit_GFX.h for display
 - WiFi.h and custom QR libraries for internet and QR generation
 - Arduino IoT Cloud: Optional platform for remote data monitoring and logging
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8. Working Principle

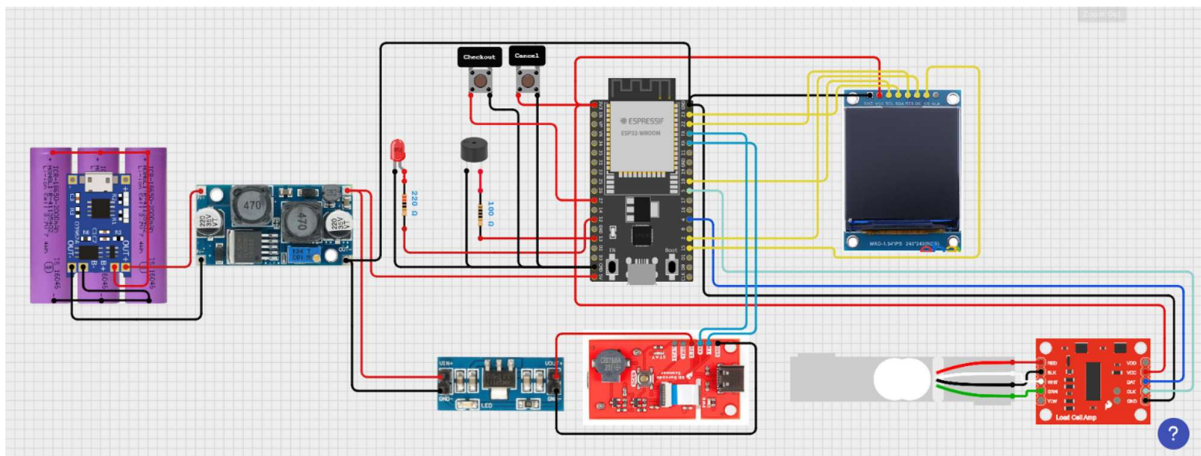
The system operates as follows:

1. Barcode is scanned and mapped to product data (name, price, expected weight).
 2. Product is placed on load cell; weight is measured.
 3. Weight is compared with the expected value.
 4. If valid, the item is added to the bill.
 5. The display shows updated billing information.
 6. User can cancel items using a dedicated button.
 7. On checkout, a payment QR code is generated.
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9. Circuit Diagram

The schematic illustrates connections between all modules:

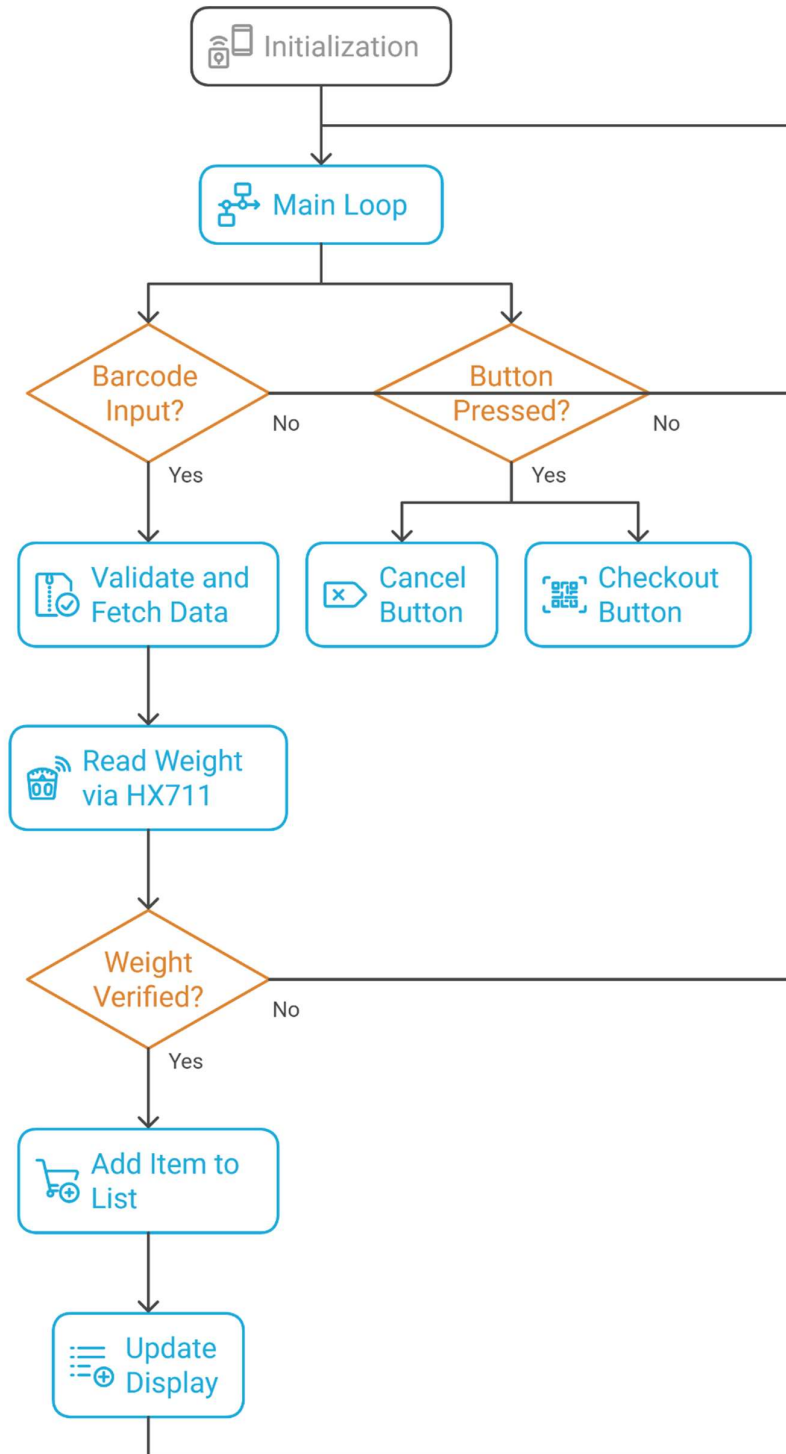
- Barcode scanner to UART pins of ESP32
- HX711 (DT, SCK) to GPIO pins
- Display to I2C/SPI pins
- Buttons to digital GPIOs
- Power Supply (Three lithium-ion batteries connected in parallel are charged using a TP4056 module with protection; the output is connected to an XL6009 boost converter set to 5V, which powers the ESP32 and a separate 3.3V regulator module used to power the barcode scanner reliably)



10. Program Flow / Logic

- Initialization: Set up serial ports, display, load cell, buttons
- Main Loop:
 - Wait for barcode input
 - Validate and fetch item data
 - Read weight via HX711
 - Verify weight, then add item to list
 - Update display with total and items
- Button Functions:
 - Cancel Button: Removes scanned item
 - Checkout Button: Generates payment QR

Retail Checkout System Flowchart



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11. Power Supply Design

To ensure stable and safe operation:

- Three lithium-ion batteries are connected in parallel to increase capacity and maintain a constant voltage.
 - The battery pack is charged using a TP4056 charging module with protection features.
 - The output of the TP4056 is connected to a boost converter set to 5V.
 - This 5V output powers both the ESP32 and a 5V to 3.3V regulator module.
 - The 3.3V regulator is used to power the barcode scanner module reliably.
 - Efficient wiring and thermal management ensure consistent performance and safety during operation.
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12. Implementation

Implementation steps:

1. Assemble hardware components on breadboard or PCB.
 2. Upload firmware developed in Arduino IDE.
 3. Test scanning, weight verification, and display logic.
 4. Calibrate the load cell.
 5. Test QR generation and UPI payment compatibility
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13. User Interface

- Display Sections:
 - Scanned items with name and price
 - Total cost summary
 - QR code for payment
 - Button Functions:
 - Cancel: Removes the scanned item from the cart and updates the display accordingly
 - Checkout: Finalizes the billing process and displays the payment QR code
 - Feedback: Optional buzzer or LEDs for scan acknowledgment, cancellation confirmation, or error notifications
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14. Applications

- Self-service billing in supermarkets
 - Unmanned microstores or vending kiosks
 - Retail automation in small shops
 - Inventory control and smart carts
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15. Advantages

- Drastically reduces checkout time
 - Increases customer autonomy
 - Low-cost and scalable solution
 - Detects mismatched item entries via weight
 - Easily customizable for various retail scales
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16. Limitations & Challenges

- Barcode database must be kept updated
 - Weighing system may need frequent calibration
 - Limited space for display on small form factor
 - Wireless payment integration requires testing and security checks
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17. Future Enhancements

- Switch from barcode to RFID for quicker scanning
 - Cloud-sync for real-time inventory management
 - Integration with Android/iOS apps
 - Biometric or face recognition for user login
 - AI-enabled fraud detection
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18. Conclusion

The Smart Billing System provides a practical, scalable, and user-friendly solution for automating the checkout process in retail spaces. By integrating embedded hardware and smart software solutions, the system significantly improves operational efficiency, minimizes human errors, and reduces overall labor costs.

This project showcases the effective implementation of IoT and embedded technologies in solving real-world problems. The integration of modules like the barcode scanner, load cell, and ESP32 microcontroller not only enhances accuracy but also introduces features such as anti-theft validation and user-friendly cancellation and payment options.

Its cost-effectiveness and modular design make it suitable for a wide range of retail formats, from small shops to large supermarkets. Additionally, the project lays the foundation for future developments like RFID integration, cloud data management, and mobile app connectivity.

Overall, the Smart Billing System is a step toward the digital transformation of retail, empowering both businesses and customers with a faster, smarter, and more secure checkout experience.

19. References

- DE2120 Barcode Scanner Datasheet
- HX711 Arduino Library Documentation
- ESP32 Official Documentation
- Arduino Project Hub Examples
- SparkFun Product Tutorials